Roll No. .....

TPE–413

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

MUD LOGGING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is E & P company ? What is the objective of it ? Write the different sectors of an E & P industry. (CO1)

OR

(b) Explain the various stages of upstream in oil and gas life cycles. (CO1)

2. (a) What is mud logging ? Why is it required ? What is the difference between mud logging and well logging ? What are its limitations ? (CO1)

OR

(b) Write the principle of mud logging. Describe different component of mud logging units. What are the various sensors available in recording the data ? (CO1)

P. T. O.

3. (a) Explain the term Rig up and rig down in mud logging. What is the precaution to be taken during this process ? (CO1)

OR

(b) How is the mud logging being useful in Formation Evaluation ? Explain Lag up, lag-Down, and complete cycle in details. (CO1)

4. (a) What is Lag time and Lag strokes ? Why is it important ? How is it calculated ? (CO2)

OR

(b) What are the various mud logging parameters recorded during data acquisition ? What are the various sensors used for recording mud logging data ? (CO2)

5. (a) Explain mud dog. How the various components of mud log to be put in various tracks for interpretation ? Draw a diagram. (CO2)

OR

(b) What are the uses of drilling cutting analysis during mud logging ? What is shale shaker and why it is required ? (CO)

Roll No. .....

TPE–402

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

RESERVOIR ENGINEERING—I

Time : 1½ Hours

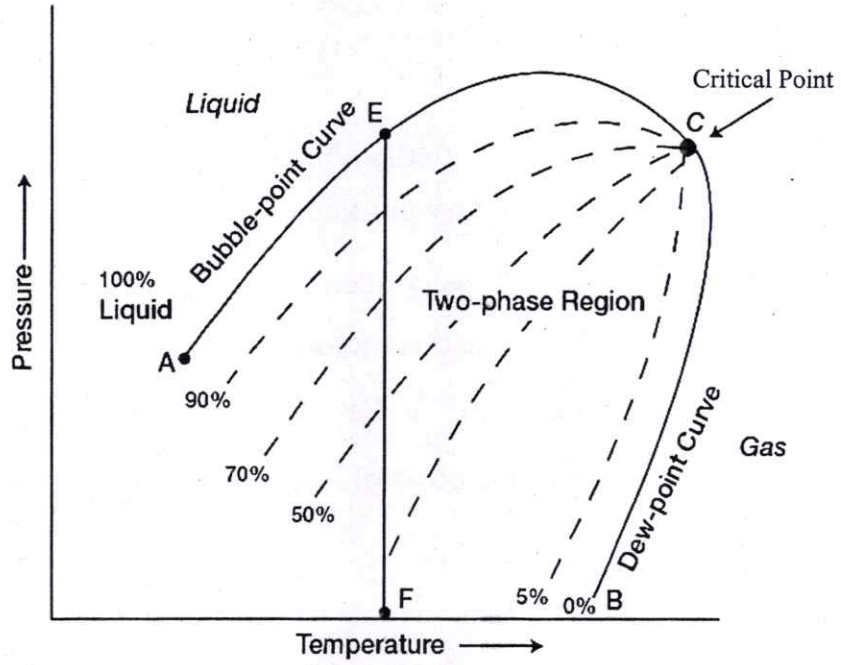
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What are the different physical properties of crude oil ? Explain. (CO1)

OR

(b) In the given figure below, explain points (C, E, F), line (AC, BC, BCA) and quality lines. (CO1)



P. T. O.

2. (a) Explain Porosity, types of porosity. What are the different factors that affect porosity ? Explain. (CO1, CO2)

OR

(b) Explain, what is a reservoir rock. What are properties should a rock have to become a good reservoir rock ? Also explain sandstone and carbonate reservoir /reservoir rock. (CO1, CO2)

3. (a) Describe structural, stratigraphic and combinations traps. (CO1, CO2)

OR

(b) Why porosity of sandstone formation does not decrease drastically in comparison to shale formation ? Explain. (CO1, CO2)

4. (a) Explain, what is a Reservoir and Reservoir Engineering and also describe difference between conventional, unconventional, oil and gas reservoir. (CO1)

OR

(b) Write short notes on the following : (CO1)

(i) Gas cap reservoir

(ii) Packing

(iii) Explain cap rock, permeability and also differentiate between primary and secondary porosity.

5. (a) Describe Near critical gas condensate reservoir, Dry gas reservoir, Wet gas reservoir and Saturated reservoir. (CO1, CO2)

OR

(b) Classify crude oil on the basis of chemical composition. (CO1, CO2)

50

Roll No. .....

TPE–403

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

HEAT AND MASS TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the laws obeys the conduction. What do you mean by the thermal conductivity of the material ? How conductivity of the material varies with temperature in solid and gas ? Derive an expression for the rate of heat transfer for the single plane wall. (CO1)

OR

(b) Derive the mathematical expression for the logarithmic mean area of the plane wall and cylindrical shape (A_m). Hence calculate the 'A_m' for A hollow cylindrical having inner and outer diameter as 15 and 25 cm, the length of the cylindrical is 100 cm. (CO1)

P. T. O.

2. (a) Two slabs, each 120 mm thick, have thermal conductivities of 14.5 W/m °C and 210 W/m °C. These are placed in contact, but due to roughness, only 30 percent of area is in contact and the gap in the remaining area is 0.025 mm thick and is filled with air. If the temperature of the hot surface is at 220°C and the outside surface of the other slab is at 30°C, determine : (CO2)

   (i) Heat flow through the composite system.

   (ii) The contact resistance and temperature drop in contact.

   Assume that the conductivity of air is 0.032 W/m °C and that half of the contact is due to either metal.

OR

   (b) What do you mean by insulation and critical radius of insulation ? Derive an expressions for the critical radius of insulation for the sphere. Specify the symbol used in the derivation. (CO2)

3. (a) Derive an expression for the heat conduction through a composite hollow cylinder considering both the conduction and convection taking place. Write the general expression for the *n*th composite cylinder. (CO2)

OR

   (b) A gas filled tube has 4 mm inside diameter and 100 cm length. The gas is heated by an electrical wire of diameter 200 microns located along the axis of the tube. Current and voltage drop against the heating element are 2 amps and 15 volts, respectively. If the measured wire and inside tube wall temperature are 400 °C and 350 °C respectively, determine the thermal conductivity of the gas filled in the tube. (CO2)

4. (a) Discuss the different types of Heat exchanger on the basis of the following :(i) nature of heat exchange process (ii) relative direction of fluid motion and (iii) design and constructional features ?

(CO1, CO2)

OR

(b) In a certain double pipe heat exchanger hot water flows at a rate of 5000 kg/h and gets cooled from 95°C to 65°C. At the same time 50000 kg/h of cooling water at 30°C enters the heat exchanger. The flow conditions are such that overall heat transfer coefficient remains constant at 2270 W/m^2 K. Determine the heat transfer area required and the effectiveness, assuming two streams are in parallel flow. Assume for both the streams $C_p = 4.2$ kJ/kg K. (CO3)

5. (a) Derive the relation of logarithmic mean temperature difference [LMTD] for parallel flow heat exchanger. Specify the symbol used. (CO2)

OR

(b) A counter flow double pipe heat exchanger using superheated steam is used to hot water at the rate of 10500 kg/h. The steam enters the heat exchanger at 180. °C and leaves at 130 °C. The inlet and exit temperature of water are 30 °C and 80 °C respectively. If the overall hear transfer coefficient from steam to water is 814 W/m. °C, calculate the heat transfer area. What would be the increase in the area if the fluid flows were parallel ? (CO2)

Roll No. .....

TMA–403

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

NUMERICAL AND GEOSTASTICAL METHODS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) The quadratic equation $2x^2 - 3x + 3 = 0$ is to be solved numerically starting with an initial guess as $x_0 = 2$. Calculate the new estimate of x after the five-iteration using Newton-Raphson method. (CO1)

OR

(b) Define absolute, relative, and percentage errors. Find the absolute, relative and percentage errors if x is rounded-off to three decimal places. Given $x = 0.005998$. (CO1)

2. (a) Find a positive real root of an equation $xe^x = 1$ by using iteration method; correct up to three decimal places between 0 and 1.

(CO1, CO3)

P. T. O.

OR

(b) A car is moving at the speed of 30 m/sec. Suddenly brakes are applied. The speed of the car per second after 't' seconds is given below :

Time (t)	Speed (v)
0	30
5	24
10	19
15	16
20	13
25	11
30	10
35	8
40	7
45	5

Using Simpson's $\frac{\text{'3'}}{8}$ rule compute the distance moved by the car in 45 seconds. (CO1,CO3)

3. (a) Using Gauss Jordan method, solve for the given set of linear equations : (CO1, CO4)

$$x - y + 3z = 9$$

$$x + y + z = 6$$

$$x - y + z = 3$$

OR

(b) Solve $\frac{dy}{dx} = xy$ numerically from $x = 0$ to $x = 1$ using explicit forward first order Euler method with initial condition of $y(0) = 1$ and step size (h) of 0.2. Find the value of $y(1)$ (correct to three decimal places). (CO1, CO4)

4. (a) Find a real root of equation : (CO1)

$$3x = \cos x + 1$$

correct to two decimal places using bisection method.

OR

(b) Solve the following equations by Gauss's elimination method : (CO1)

$$2x + 4y + z = 3$$

$$3x + 2y - 2z = -2$$

$$x - y + z = 6$$

5. (a) A curve is drawn to pass through the points given by the following table : (CO3, CO4)

x	$f(x)$
1	2
1.5	2.4
2.0	2.7
2.5	2.8
3.0	3
3.5	2.6
4.0	2.1

Find the value of the integral $\int_1^4 f(x)\,dx$ using Simpson's rule.

OR

(b) Consider an initial value problem (IVP) $\frac{du}{dt} = 3t^2 + 1$ with $\mu(0) = 0$.

Given IVP solved by numerically using Runge-Kutta method calculate $\mu(0.2)$ taking step-size $(h = 0.1)$. (CO3, CO4)

50

Roll No. .....

TPE–401

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

PETROLEUM GEOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Describe source rock, categories and types of source rock, and explain depositional environments in which they are typically deposited. (CO1)

OR

(b) Crude oil is composed of various Chemical constituents/compounds. Describe those chemical constituents/compounds. (CO1)

2. (a) Write short notes on the following : (CO1)

(i) Inorganic theory of origin of hydrocarbons

(ii) Lacustine and Carbonate reservoir rocks

P. T. O.

OR

(b) Describe Sedimentary Basins of India also write all sedimentary basin of India (categorywise). (CO1, CO3)

3. (a) Define Reservoir Traps and what are different types of reservoir traps. With the help of a neat sketch explain stratigraphic traps. (CO)

OR

(b) (i) Draw structures of the following with their boiling point :

– 2, 2 Di Methyl Butane

– 2, 3 Di Methyl Butane

– Cycloheptane

– Ethane

– Isobutane

(ii) A well produces 380 m^3 of oil. The oil density is 750 kg/m^3. Determine specific gravity and API Gravity. (CO1, CO3)

4. (a) Define pour point of crude ? What are the problems associated with high pour point crude ? How to minimize these problems. (CO2)

OR

(b) Describe all physical properties of crude oil. (CO1, CO3)

5. (a) Explain the concept of Migration of Hydrocarbon and also describe Primary and Secondary Migration. (CO1, CO3)

OR

(b) Describe the following : (CO1, CO3)

(i) Sandstone Reservoir

(ii) Clastic Rock

(iii) Non-Clastic Rock

(iv) Specific Gravity

(v) Biotic Theory

Roll No. .....

TPE–405

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

PETROLEUM DRILLING ENGINEERING—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are the objectives of well planning ? When planning for a vertical well, several parameters need to be considered to ensure successful drilling and completion. Describe, what are those parameters to be considered in well planning for a vertical well. (CO1)

OR

(b) List and describe the types of well drilling methods used in oil and gas industry. (CO1)

2. (a) List and describe the component of hosting system of a rotary drilling rig. (CO1)

OR

(b) List and describe the component of rotary system of a rotary drilling rig. (CO1)

P. T. O.

3. (a) List five major causes of kicks. What is the typical sequence of events that may be observed if a kick is taken during a connection ? What information is required prior to initiating kill procedures ? (CO2)

OR

(b) Calculate the liner size required for double-acting duplex pump where rod diameter is 2.5 in, stroke length is 22 in stroke, pump speed is 70 strokes/min. In addition, the maximum available pump hydraulic horsepower is 1200 hp. For optimum hydraulics, the pump recommended delivery pressure is 3,000 psi. Assume the volumetric efficiency of pump is 98%. (CO2)

4. (a) Drilling fluid plays a critical role in ensuring the success, efficiency, and safety of drilling operations. Describe the key role of drilling fluids in drilling operations of oil and gas wells. (CO3)

OR

(b) Give the classification of drilling fluid on the basis of Base Fluid, Rheological Properties and Additives. (CO3)

5. (a) Discuss the key steps involved in design of drilling fluid for an oil and gas well. (CO3)

OR

(b) Discuss the common additives used in drilling fluids and their functions. (CO3)

50

Roll No. .....

TPE–404

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

APPLIED SEDIMENTOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you understand by weathering ? Discuss the different types of mechanical weathering. (CO1)

OR

(b) Describe the processes that are involve in digenesis and explain how its effects the reservoir quality. (CO1)

2. (a) Discuss about various types of permeability and analyze their significance in the field of petroleum exploration. (CO1)

OR

(b) Define porosity. Discuss about the various factors that affect porosity. (CO2)

3. (a) Describe the different sedimentary rock textures in detail. (CO1)

P. T. O.

OR

(b) Differentiate between : (CO2)

(i) Abrasion and Exfoliation

(ii) Laminar and Turbulent flow

4. (a) Explain the following : (CO1)

(i) Clastic and non-clastic sediments

(ii) Wentworth grade scale

OR

(b) Write a short notes on the following : (CO2)

(i) Roundness and spherecity

(ii) Reynolds' Numbers

5. (a) Describe in detail about the agents involved in the transportation of sediments. (CO1)

OR

(b) Discuss the role of fluids density and viscosity in sediment transportation. (CO1)